

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (Mechanical) (Credit System) Degree Examinations**

January 2023

19ME702 - MECHATRONICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) What is mechatronics? Brief evolution of mechatronics.	10	L*2	1	1
	b) Write a short note on i) Eddy current sensor ii) Hall effect sensor	10	L2	1	1
2.	a) Explain how microprocessors are used in I.C. Engine Management system with a block diagram.	10	L2	1	1
	b) Define transducers. Sketch and explain capacitive transducer.	10	L2	1	1
Unit – II					
3.	a) Sketch and explain the working principle of pneumatic systems.	10	L2	2	1
	b) What is direction control valve? Explain the operations of a single solenoid valve.	10	L2	2	1
4.	a) Sketch and explain: (i) air filter (ii) OR valve.	10	L2	2	1
	b) Explain the pneumatic application circuit w.r.t. time delay.	10	L3	2	2
Unit – III					
5.	a) List different types of solid state switches and explain transistor characteristics.	10	L3	2	2
	b) Explain the following Motors. i) Variable reluctance stepper Motor ii) Hybrid stepper Motor	10	L2	3	1
6.	a) Explain thyristor circuit and thyristor characteristics.	10	L2	3	1
	b) Explain basic principle, working and parts of a D.C motor with a neat sketch.	10	L2	3	1
Unit – IV					
7.	a) Define signal conditioning, what are the necessity for signal conditioning.	04	L2	4	1
	b) Explain balance mode of Wheatstone bridge and hence deduce the expression for change in output voltage.	10	L2	4	1
	c) With block diagram explain data acquisition system.	06	L2	4	1
8.	a) Derive voltage out equation for inverting amplifier and differential amplifier.	12	L3	5	1
	b) Explain the principle of ADC signal.	08	L2	5	1
Unit – V					
9.	a) Explain the following terminologies related to microprocessor. i) Program counter ii) Flag register iii) Stack pointer	10	L2	5	1
	b) Explain with a neat sketch, the internal architecture of INTEL 8085 microprocessor.	10	L2	5	1
10.	a) Define programmable logic controller. Briefly explain external structure of PLC.	10	L2	5	1
	b) Explain the basic standard symbols used in ladder diagram for programming PLC.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
